Structured Programming Lab

Bachelor Program Business Informatics

Department of Computer Science

Formatted Input

Reading Integer Values

Reading Integer Values - Code

```
1  package FormattedInput;
2
3  import java.util.Scanner;
4
5  public class ReadingIntegerValuesDemo {
6
7      public static void main(String[] args) {
8          Scanner sc = new Scanner(System.in);
9          System.out.printf(": ");     //prompt for input
10         int time = sc.nextInt();     //read an integer value
11         System.out.printf("time = %d\n", time); //print it again
12         sc.close();
13     }
14
15 }
```

Reading Integer Values - Output

(text in red is user input)

```
: 1245
time = 1245
```

Reading Integer Values - Explanation I

- A `Scanner` is used to read input from `System.in`
- `System.in` is a *stream* from which the program *reads* input. For now, think of it as the keyboard.
- The `Scanner` *class* is *imported*, i.e. it is made available in our program.

```
3 import java.util.Scanner;
```

- A `Scanner` object named `sc` is created and connected to `System.in`

```
8         Scanner sc = new Scanner(System.in);
```

- The `Scanner` is instructed to read an integer value (which is *stored* in a *variable* named `time`)

Reading Integer Values - Explanation II

```
10          int time = sc.nextInt();     //read an integer value
```

- The *variable* named `time` now *holds* the value that was entered
- the *value* of variable `time` is printed

```
11          System.out.printf("time = %d\n", time); //print it again
```

- The `Scanner` is closed at the end of the program

```
12          sc.close();
```

Reading Double Values

Reading Double Values - Code

```
1  package FormattedInput;
2
3  import java.util.Scanner;
4
5  public class ReadingDoubleValuesDemo {
6
7      public static void main(String[] args) {
8          Scanner sc = new Scanner(System.in);
9          System.out.printf(": ");
10         double temperature = sc.nextDouble();
11         System.out.printf("temperature = %.1f%n", temperature); //print it
12         sc.close();
13     }
14
15 }
```

Reading Double Values - Console

```
: 22.8
temperature = 22.8
```

Reading Double Values - Explanation I

- The `Scanner` is instructed to read a **`double`** value (which is *stored* in a *variable* named `temperature`)
- the variable is of type `double`

```
10          double temperature = sc.nextDouble();
```

Reading Strings

Reading Strings - Code

```java
package FormattedInput;

import java.util.Scanner;

public class ReadingStringsDemo {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.printf(": ");
        String lastname = sc.next();
        String firstname = sc.next();
        String greeting = sc.nextLine();
        System.out.printf("%s %s %s\n", greeting, firstname, lastname);
        sc.close();
    }

}
```

Reading Strings - Console

```
: Mustermann Max Guten Morgen
Guten Morgen Max Mustermann
```

Reading Strings - Explanation I

- with sc.next() everthing till the next space or line break is read
- with sc.nextLine() a complete line is read (everthing till the next line break)
- it is read as a String

```
10          String lastname = sc.next();
11          String firstname = sc.next();
12          String greeting = sc.next();
```

Reading Strings - MiniExercise

Reading Strings - PersonDetails

Read the following details of a person on the console

- firstname
- lastname
- date of birth
  - year
  - month
  - day
- weight (in kg)
- height (in m)

and pretty-print them on the console (names are truncated to a max of 7 characters, height and weight are displayed with one digit after the decimal separator)

Example:

```
: Arnold Schwarzenegger 1947 7 30 113.5 1.88
Arnold Schwarz 30.07.1947 1.9m 113.5kg
```